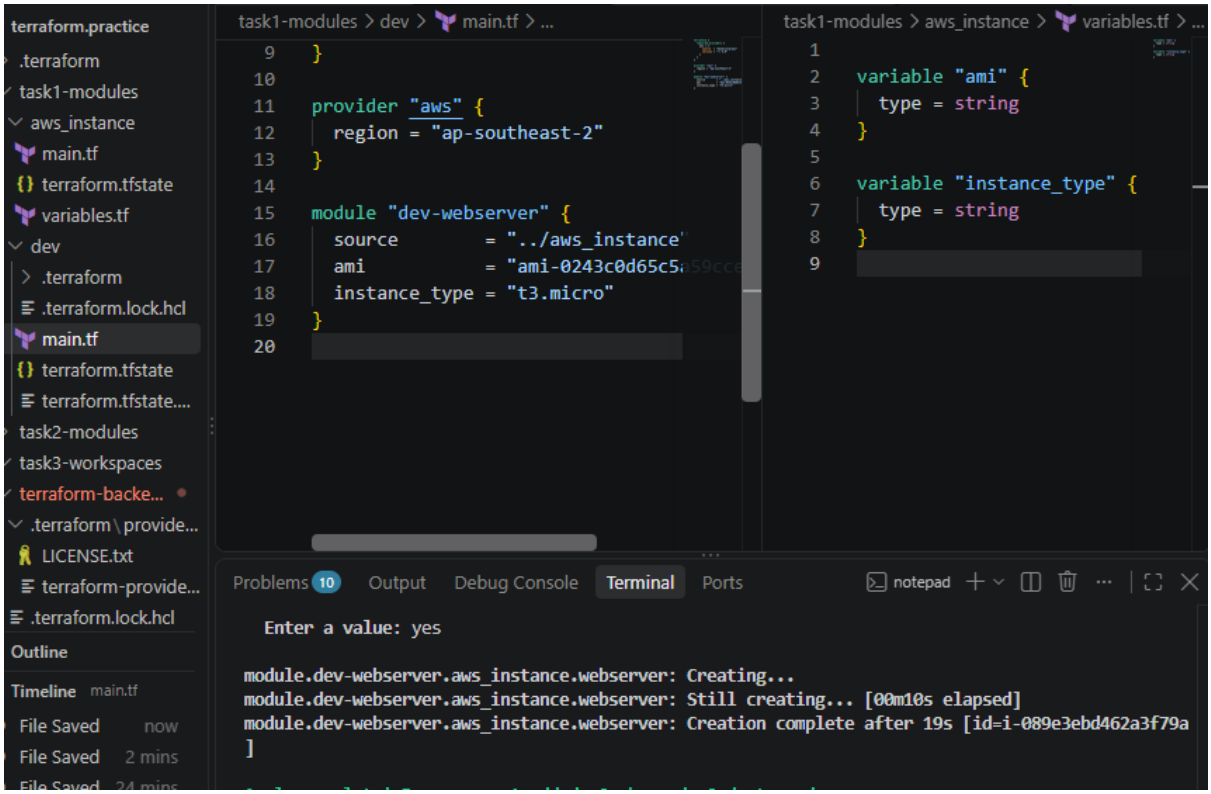
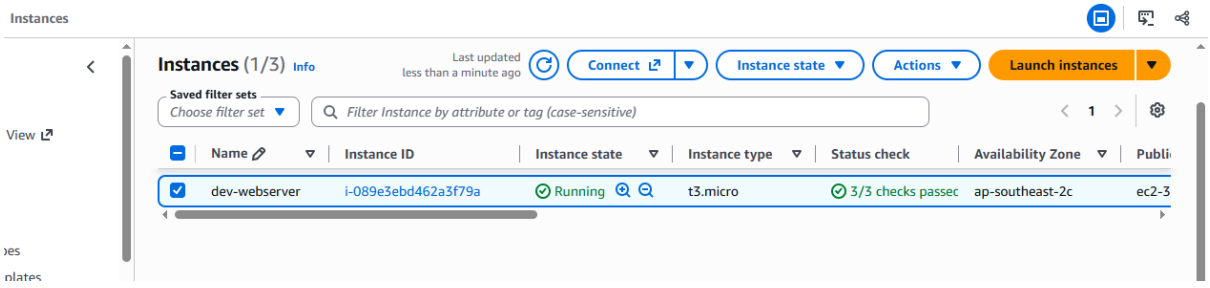
Created EC2 using Terraform Modules

1. Created a reusable EC2 module using Terraform.
2. Created a separate directory for EC2 module configuration.
3. Defined EC2 resources using the `aws_instance` resource.
4. Created input variables for AMI ID and instance type.
5. Used the `module` block to call the child module.
6. Passed AMI ID and instance type from the Dev environment.
7. Configured AWS provider in the Sydney region.
8. Executed Terraform initialization, validation, planning and applied the configuration.
9. Resolved AMI ID and AWS permission issues.

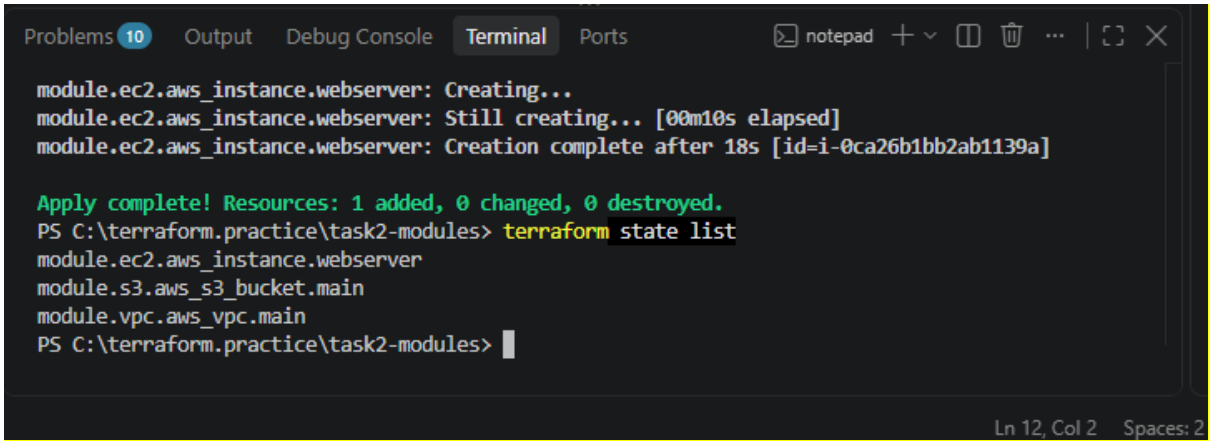




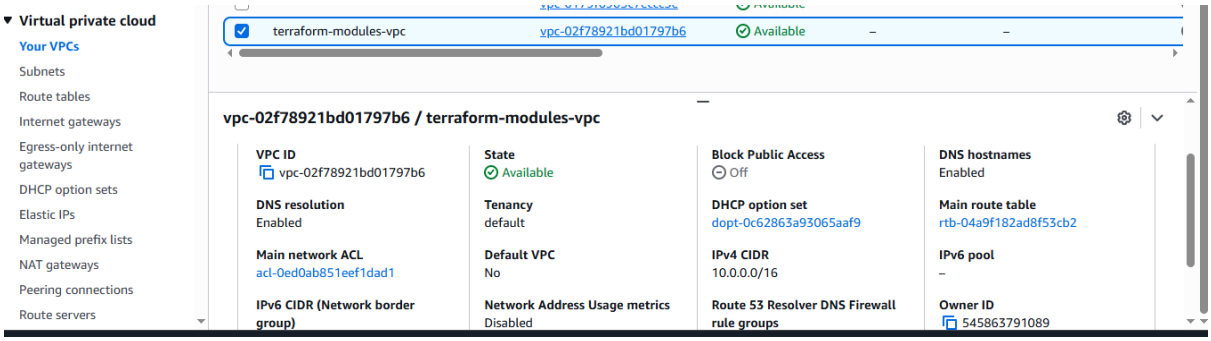
Provision EC2, S3, and VPC using Terraform Modules

Documentation Points

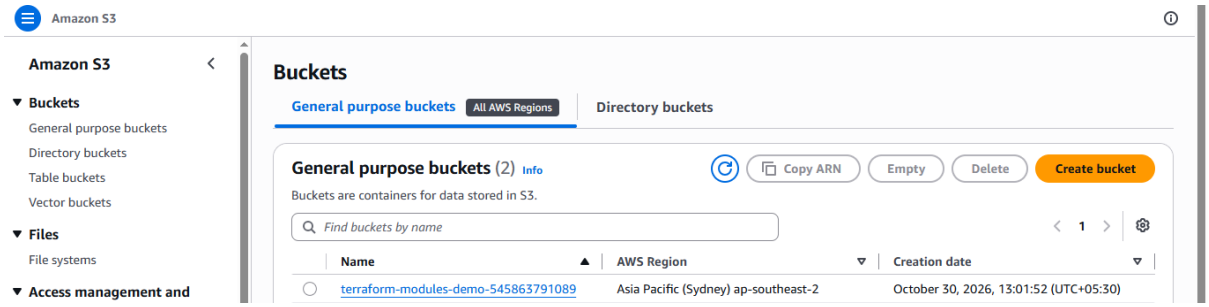
1. Created separate Terraform modules for EC2, S3, and VPC.
2. Configured AWS provider with region `ap-southeast-2`.
3. Created reusable module configurations using separate `main.tf` files.
4. Defined input variables for EC2 configuration.
5. Called the EC2, S3, and VPC modules from the root `main.tf` file.
6. Configured the Amazon Linux AMI ID and EC2 instance type.
7. Initialized Terraform using `terraform init`.
8. Validated the configuration using `terraform validate`.
9. Reviewed resource creation using `terraform plan`.
10. Verified Terraform-managed resources using `terraform state list` and AWS CLI commands.



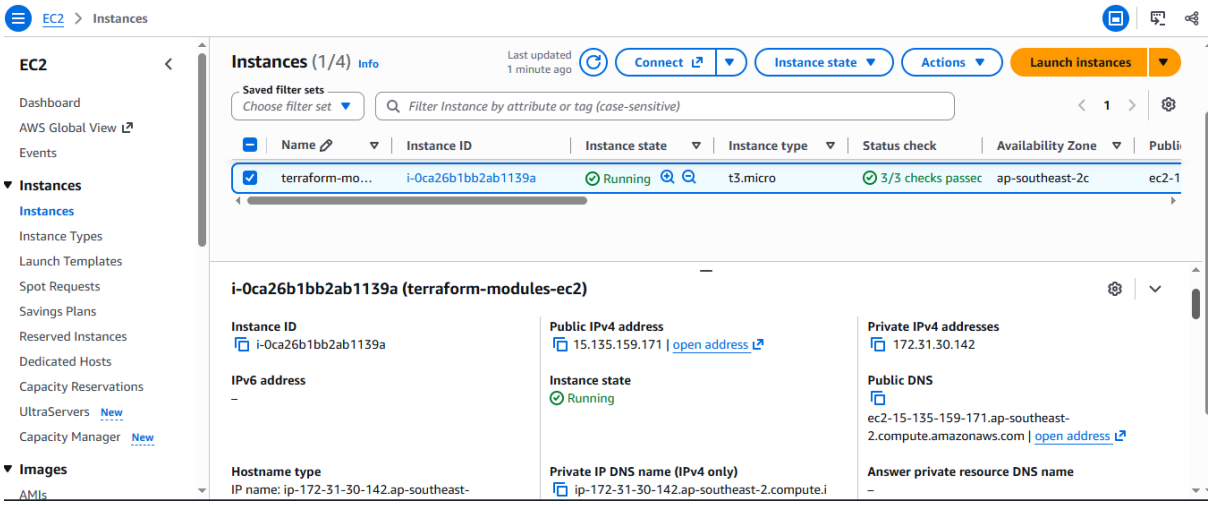
VPC



S3



EC2



Terraform Workspaces – Completed

Documentation Points

1. Created a Terraform project for multiple environments.
2. Configured the AWS provider in the Sydney region (`ap-southeast-2`).
3. Retrieved the Amazon Linux 2023 AMI using the AWS AMI data source.
4. Created an EC2 instance resource using Terraform.
5. Used `terraform.workspace` to identify the active environment.
6. Used local values to define environment-specific EC2 instance types.
7. Configured environment-specific EC2 tags and names.
8. Created separate Terraform workspaces for Dev, Staging, and Prod.
9. Used `terraform workspace list` to verify available workspaces.
10. Used `terraform workspace select` to switch between environments.
11. Executed `terraform plan` to review infrastructure changes for each workspace.

terraform workspace list
terraform workspace show
terraform state list
terraform plan

